

Exploring adsorption and sensing mechanism of Li-decorated N-doped black phosphorene for toxic gases: Insights from first-principles calculations

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ARTICLE INFO

Keywords:

First-principles calculations
Metal-decorated phosphorene
Toxic gases
Coverage
Adsorption

ABSTRACT

The two-dimensional material black phosphorene has attracted extensive attention due to its unique sensing performance. Herein, on the basis of first-principles calculations, we investigate the adsorption and dissociation of several toxic gases (NO₂, SO₂, H₂S) on the Li-decorated N-doped black phosphorene (LNBP). The effect of gas molecule coverage on the electronic and sensing properties of LNBP is also studied. Our results demonstrate that NO₂ and SO₂ have a strong interaction with LNBP due to the hybridization of O-2p with P-3p orbitals, while the 3NO₂-LNBP system exhibits a high charge transfer up to 0.95 e from the Li atom to gas molecules. Furthermore, the capture ability of toxic gas on the LNBP follows the sequence of H₂S < SO₂ < NO₂, which corresponds with their adsorption energy of -0.68 eV, -1.37 eV and -2.55 eV. In addition, the LNBP is appropriate for NO₂ gas detection with a high sensitivity ranging from 84.8 % to 92.6 % as the coverage increases. The significant changes in the work function of NO₂ and SO₂ suggest a high selectivity in LNBP. The energy barrier of dissociation is relatively low at 1.24 eV with an exothermic process for the NO₂-LNBP system. We believe that these findings provide an important insight into sensing applications of metal-decorated black phosphorene.

1. Introduction

When toxic gases are released into the workplace by chemical industrial production and automobile exhaust emission, they give rise to a plethora of security concerns ranging from environmental pollution to human health [1,2]. Among them, NO₂, SO₂, and H₂S are widely publicized for their harmful effects, such as damaging workers' eyes and inducing respiratory inflammation. In addition, NO₂ and SO₂ readily dissolve in the water droplets in clouds, causing acid rain that affects the natural balance of rivers, lakes, and soils, resulting in damage to wildlife and vegetation [3-5]. To address these challenges, it is imperative to develop materials capable of effectively absorbing and precisely detecting the toxic gases during chemical production [6].

Over recent years, various types of materials have been studied for this purpose, including metal oxide semiconductors and polymers [7,8]. Remarkably, there are widespread researches on the utilization of two-dimensional (2D) materials for gas sensing due to their inherent advantages, which include a higher surface-area-to-volume ratio, lower electrical noise, and tunable band structure [9-11]. Among those 2D materials, the few-layered black phosphorene (BP) fabricated through mechanical exfoliation manifests a suitable band gap of 1.72 eV and a

high room-temperature carrier mobility of up to 1000 cm² V⁻¹ s⁻¹, holding great promise for various fields [12-14]. Many reports also observe that phosphorene outperforms graphene and MoS₂ in terms of adsorption for gas molecules because of its particular selectivity [15]. Both theoretical and experimental studies demonstrate excellent gas sensitivity of BP [16-20]. For instance, Abbas et al. prove the prominent performance of BP in detecting NO₂ molecules down to 5 ppb at room temperature, outperforming that of other 2D materials [18]. Additionally, it can be made as a field-effect transistor based on BP nanosheet to exhibit ultrahigh sensitivity for toxic gas at high adsorption density by Cui et al. [19]. However, there is evidence to suggest that the adsorption strength of small gas molecules on pristine BP is affected due to its fast degradation [21,22].

Among various improvement strategies, introducing doping atoms into BP better increase the chemical activity and stability of P atoms [23]. As one of the most common dopants, nitrogen (N) has a similar electronic structure to phosphorus, which allows BP to be maintained originally with minimal strain [24]. Several investigations provide pieces of evidence that N-doped phosphorene (NBP) serve as an alternative approach to simultaneously improve its electron mobility, with moderate adsorption energy and substantial charge transfer suitable to

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<https://doi.org/10.1016/j.surfin.2024.104872>

Received 12 April 2024; Received in revised form 7 July 2024; Accepted 26 July 2024

Available online 27 July 2024

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detect gases [25,26]. Meanwhile, the NBP nanosheets have been synthesized experimentally, which paves the way for further exploration of the sensing potential of NBP-based materials using theoretical simulations [27,28].

It is widely accepted that metal decoration enhances the gas sensitivity of nanomaterials, in which the metal atoms play an important role in promoting the electron mobility of the nano-surface and thus increase the charge transfer of the gas adsorption systems. Therefore, they conduct substantial theoretical and experimental work to assess the gas-sensing performance of metal-decorated 2D materials [29–33]. The puckered surface morphology of BP offers a larger adsorption space for metal atom decoration engineering, and the introduction of metal atoms makes it easier for unoccupied orbitals to tune the surface states of the substrate for gas adsorption. It is reported that transition metals effectively enhance the adsorption abilities of 2D materials but are limited in practical applications due to larger cohesive energy and atomic mass [34]. Therefore, alkali and alkali earth metals with smaller sizes and lighter weights are investigated to decorate BP. For example, the hydrogen storage capacity of the phosphorene is effectively strengthened through lithium (Li) decoration on its surface by Yu and Halder et al. [35,36]. From the toxic gas detection, the adsorption of sulfur-based and nitrogen-based gas molecules on various metal-decorated phosphorene is investigated systematically by Lei et al. [37,38]. It is noteworthy that the adsorption strength of phosphorene is improved by Li modifiers, displaying interesting magnetism and electrical transitions with great potential for gas detection. Kuang et al. examine the adsorption and dissociation of several acidic gases on metal-modified phosphorene, revealing that light metals (Li, Al) show stronger adsorption compared to transition metals [39]. Marjani et al. utilize the Li-decorated BP (LBP) for capturing and detecting nitrogen dioxide and find that LBP demonstrates significant efficiency in detection at 79 % [40].

Regretfully, most reports about NBP focus on its electrocatalytic and electronic properties, and its mechanism of gas adsorption, especially the metal-decorated configurations, are under limited investigation. Herein, the adsorption behavior and sensing properties of Li-decorated N-doped black phosphorene (LNBP) towards toxic gases (NO_2 , SO_2 , H_2S) at three coverages are investigated by employing first-principles calculations based on density functional theory. Besides, the dissociation of these toxic gas molecules is also studied to assess their catalytic ability. These theoretical studies help us better understand the mechanism, offering valuable perspectives into utilizing metal atom-decorated phosphorene for simultaneous monitoring and desorption of toxic gases.

2. Computational details

All first-principles calculations were carried out by using the Cambridge Sequential Total Energy Package (CASTEP) based on density functional theory [41]. The exchange-related energy was calculated using the generalized approximate gradient (GGA) and Perdew-Burke-Ernzerhof (PBE) functional [42]. According to some calculated results about BP [21,43], DFT-D2 was considered to properly describe long-range forces such as van der Waals dispersion forces between BP and gas molecules [44]. Due to the underestimation of the band gap by GGA-PBE functional, the Heyd-Scuseria-Ernzerhof functional (HSE06) was also used to obtain a precise band gap [45]. A series of tests showed that an $8 \times 8 \times 1$ Monkhorst-Pack mesh and a cut-off energy of 450 eV gave well-converged results for our self-consistent calculations. All the structure optimizations were performed until the residual forces on unconstrained atoms were less than $0.01 \text{ eV}/\text{\AA}$, and the convergence criteria of the total energy was set at 10^{-5} eV . The dipole correction was considered for surface supercell calculations in this work [46].

To confirm the stability of the NBP, the formation energy (E_f) was evaluated by the formula:

$$E_f = E_{\text{doped-BP}} - E_{\text{BP}} + nE_P - E_N \quad (1)$$

Where $E_{\text{doped-BP}}$ and E_{BP} were the total energies of the doped and the pristine BP monolayers, respectively, while E_P and E_N were the chemical potentials of the host P atom in pristine BP and the dopant N atom, respectively. n is the number of P atoms being substituted.

The binding energy (E_b) was used to evaluate the stability of the NBP after Li decoration. The lower the E_b was, the more stable the material was, making it more likely to be fabricated in experiment, which were calculated by:

$$E_b = E_{\text{LNBP}} - E_{\text{NBP}} - E_{\text{Li}} \quad (2)$$

Here E_{LNBP} was the total energy of the system for LNBP, E_{NBP} was the energy of the NBP and E_{Li} was the energy of Li. To measure the adsorption ability of gas molecules on LNBP, the adsorption energy (E_{ads}) was calculated as:

$$E_{\text{ads}} = (E_{\text{total}} - E_{\text{LNBP}} - nE_{\text{gas}}) \quad (3)$$

Where E_{total} , E_{LNBP} , E_{gas} represented the energy of LNBP with absorbed gas molecules, LNBP, and absorbed gas molecules. n is the number of gas molecules being absorbed.

3. Results and discussion

3.1. Stable construction and electronic properties of LNBP

To prevent the interaction between periodic adsorbates, the 3×3 supercells are utilized, with a vacuum of $15 \text{ \AA}$ applied perpendicular to the phosphorene layer. After structural relaxation, the lattice constants of the pristine BP are $a = 4.61 \text{ \AA}$ and $b = 3.31 \text{ \AA}$, matching values reported in previous studies [47,48]. In this work, doping is the replacement of a P atom in the phosphorene by an N atom, and the decoration is mainly achieved by placing a metal atom on the surface of a substrate. The structures and geometrical parameters of both BP and NBP are displayed in Fig. 1(a, b) and Table 1. It is observed that doping an N atom results in a decrease in both interlayer distance ($1.83 \text{ \AA}$) and angle (101.46°), while the plane angle increases from 96.02° to 104.01° . Because of its distinct atomic radius, the position of the N atom decreases compared to other P atoms. Moreover, the E_f of 0.70 eV is calculated for NBP during doping, assuring the low stability in line with the previous study [25]. The band structures of BP and NBP are illustrated in Fig. S1, with band gaps of 1.46 eV and 1.52 eV . The results indicate that the doping N atom does not affect the p-type conducting properties of BP but impacts the CBM, which causes an upward shift of the conduction band to a higher energy and an increase in the band gap.

As shown in Fig 1(a), three stable decoration sites are considered at a hollow (H) site over the center of the hexagon, above the midpoint of the covalent bond (B), and at the top of a P atom (T). The E_b of LNBP at different initial adsorption distances ($2.10 \text{ \AA}$, $2.70 \text{ \AA}$, $3.50 \text{ \AA}$) are calculated to mitigate the occurrence of local minima, then the distance of $2.10 \text{ \AA}$ is selected as the final initial adsorption distance to optimize. The E_b , charge transfer (Q), and vertical height (L_v) for the decoration of Li on NBP and BP are displayed in Table S1. The E_b of all aforementioned sites are lower than the cohesion energy of lithium bulk metal (-1.63 eV), suggesting a strong interaction between the Li atom and NBP, which effectively prevents the metal cluster [36,49]. Notably, the Li atom exhibits enhanced stability when adsorbed on NBP at the T site, with an E_b and Q of -3.56 eV and 0.77 e higher than other sites. By comparison, Li decoration at the H site of BP (LBP) is constructed with an E_b of -2.20 eV , a Q of 0.7 e , and a L_v of $1.53 \text{ \AA}$ (Table S1), further confirming that NBP shows a stronger interaction with Li atom. To further determine the stability of LNBP, the phonon band structure and molecular dynamics (MD) simulation in NVT ensemble are utilized at room temperature (298 K), as shown in Fig. S3. The E_b of the LNBP does not oscillate more than 0.15 eV within 500 ps , and its positions are slightly perturbed at 100 ,

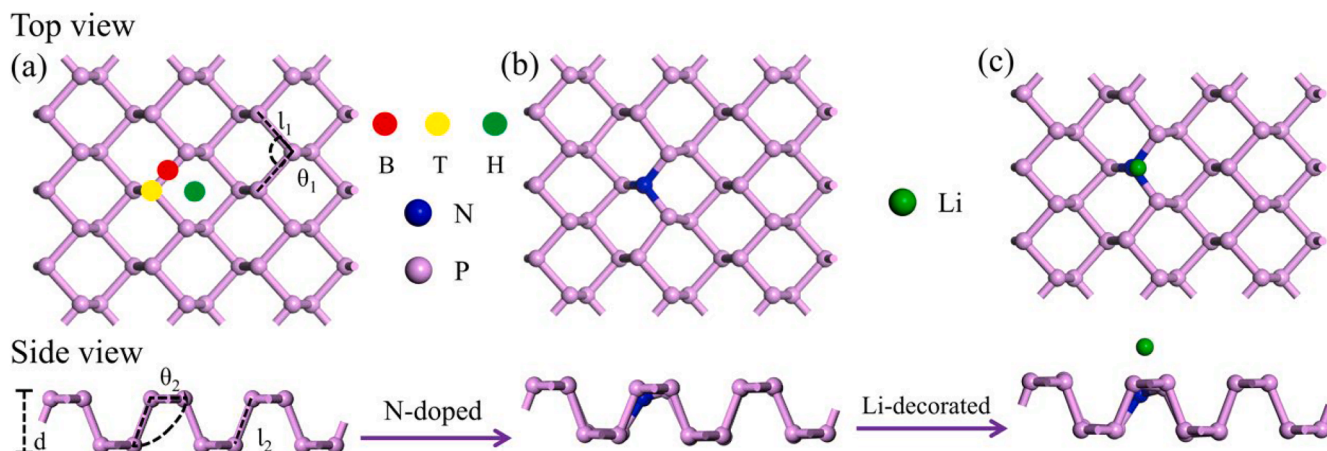


Fig. 1. Optimized structures of monolayer (a) black phosphorene, (b) N-doped black phosphorene, and (c) Li-decorated N-doped black phosphorene.

Table 1

The parameters for pristine BP and NBP. The formation energy (E_f in eV), distances between nearly P atoms and vertical distance (l_1 , l_2 and d in Å), angles included exterior and interior layer (θ_1 , θ_2 in $^\circ$), the energy gap (E_{gap} in eV).

Type	l_1	l_2	θ_1	θ_2	d	E_{gap}	E_f
BP	2.22	2.26	96.02	104.01	2.11	1.46	
NBP	2.30	1.83	104.96	101.46	2.08	1.52	0.70

250, and 500 ps near the T site. Additionally, the phonon band structure adopts the Brillouin zone consistent with the energy band, and the minimum value of the acoustic branch is clearly at zero. These results verify the thermodynamic and kinetic stability of LNBP at room temperature, as well as the possibility of successful preparation in experiments.

To investigate the mechanism underlying the influence of doping N atom on the decoration strength of Li atom, the partial density of states (PDOS), charge density difference, and band structure of LNBP and LBP are calculated respectively, as shown in Fig. 2 and Fig. S2-S3. It is observed that an impurity energy level is incorporated close to the Fermi energy, leading to a reduction in the energy gap ($E_{gap} = 1.17$ eV) and a transition from p-type to n-type semiconductor. As depicted in Fig. S2 (c), the yellow represents the region of charge aggregation, while the

blue indicates the charge depletion. The Li atom loses a total of 0.77 e of electrons, which are subsequently transferred to the NBP, with three atoms involved in the interaction in a small region, forming a charge accumulation. The distance between the Li atom and the nearest atom is Li-N (2.12 Å) and Li-P (2.53 Å and 2.54 Å), which is close to the sum of their covalent radii (2.01 Å and 2.44 Å), signifying that the existence of enhanced interaction between the Li atom and NBP. Besides, the efficient charge transfers produce strong bond energies so that the Li atom is chemically adsorbed on the surface of NBP by the formation of ionic bonds in the form of electrostatic binding forces. In comparison, the LBP mainly interacts with three neighboring P atoms at the H site with a total charge transfer of 0.70 e, indicating that the narrower range of interaction induces a more pronounced charge transfer of LNBP. The majority of the electronic density in LNBP and LBP is contributed by P-3p orbitals. The Li decoration in LNBP leads to an orbital hybridization of Li-2 s, N-2p, and P-3 s orbitals in the vicinity of the Fermi energy, generating a new coupling peak at the impurity energy level accommodating the electrons from the Li atom, resulting in an increased of Fermi energy.

3.2. Adsorption behavior of toxic gas (NO_2 , SO_2 , H_2S) on LNBP

Several common industrial gases, including NO_2 , SO_2 , and H_2S , are

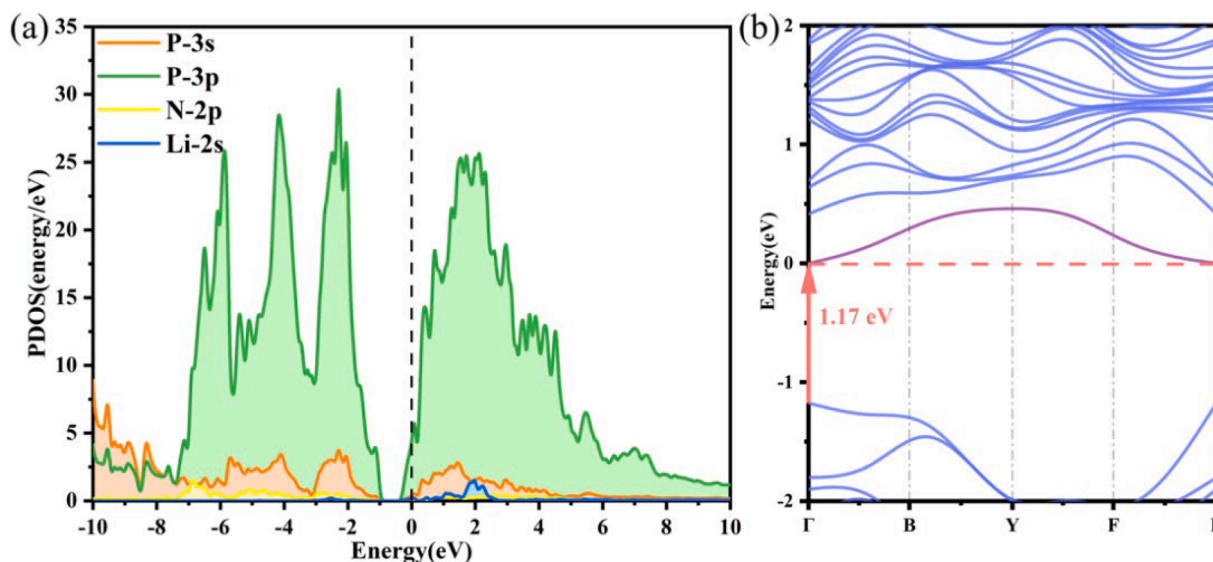


Fig. 2. The (a) partial density of states and (b) band structure of LNBP.

selected [50], and their optimized structures and geometrical parameters are presented in Fig. S4. The different placement of gas molecules will give different results. For the adsorption of NO_2 , there are four possible initial configurations of NO_2 molecule reacting with the adsorption sites (Fig. S5) as type 1–4: NO_2 is perpendicular to the surface with O–N–O angle upwards with N approaching the metal atom; NO_2 is perpendicular to the surface with single O approaching the metal atom; NO_2 is perpendicular to the surface along the zigzag direction with O approaching the metal atom, and NO_2 is perpendicular to the surface with O–N–O angle downward with O approaching the metal atom. For the adsorption of SO_2 and H_2S , due to their similar similarly connection to each other atomically, the angles of O–N–O, O–S–O, and H–S–H are obtuse. Therefore, four possible adsorption structures will be constructed separately using a similar placement as NO_2 , as shown in Fig. S6–S7. The total energy calculations with full geometry optimization are carried out for all the adsorption configurations, as shown in Table S2. By comparing the E_{ads} and the adsorption distance (d_1), the most stable structures and E_{ads} of the single gas-LNBP systems are selected to display in Fig. 3 and Table 3. The results indicate that the NO_2 molecule adsorbs above the Li atom in a direction parallel to the armchair with an E_{ads} of -2.55 eV and a d_1 of 1.92 Å. The optimal adsorption site of SO_2 on LNBP is similar to that of NO_2 , showing a d_1 of 2.03 Å and an E_{ads} of -1.37 eV. However, H_2S has weak adsorption on LNBP compared to other gases, with an E_{ads} of -0.68 eV and a higher d_1 of 2.54 Å, and the adsorption position of H_2S differs significantly. Compared to other 2D materials (Table 2), LNBP shows a better adsorption effect on these gases.

To further investigate the mechanism and selectivity of toxic gases on LNBP in a single gas-LNBP system, the total density of states (TDOS), PDOS, charge density difference, and charge transfer are displayed in Fig. 4 and Fig. 5. All the gases act as an acceptor and receive electrons from Li atom. Regarding NO_2 adsorption in Fig. 4(a), NBP starts to lose electrons and transfers to the Li atom, while NO_2 receives electrons from the Li atom to form a charge accumulation of about 0.72 e. Due to the high charge transfer between NO_2 and LNBP, strong chemisorption over the surface of LNBP is observed. The charge density difference of SO_2 adsorbed by LNBP is shown in Fig. 4(b), indicating a total of 0.46 e charge transfer during the process. More electrons are transferred similarly to the pattern of NO_2 between the Li atom and SO_2 . In comparison, the greater electronegativity of S induces stronger bonding phenomena. Fig. 4(c) depicts the adsorption of H_2S on LNBP, which

Table 2

Adsorption properties of different metal-decorated 2D materials for NO_2 , SO_2 , and H_2S molecules are compared in this paper.

System	E_{ads} (eV)	Q (e)	Ref.
NO_2 -Ag-In $_2$ O $_3$ (110)	-2.34	-0.33	[51]
NO_2 /Pd-BC $_6$ N	-1.82	0.17	[52]
NO_2 /Pt-Borophene	-1.74	-0.43	[53]
NO_2 /Ag $_3$ -WSe $_2$	-1.95	0.43	[54]
NO_2 -Pd-C $_3$ N	-2.74	-0.41	[55]
NO_2 -Rh-MoTe $_2$	-1.95	-0.29	[56]
SO_2 /Co-Pc	-0.61	-0.08	[57]
SO_2 /Ga-Silicene	-0.82	-0.18	[58]
SO_2 -Ru-InSe	-0.94	-0.002	[59]
SO_2 -Pd-G-O	-1.24	-0.07	[60]

Table 3

The adsorption energy (E_{ads} in eV), magnetic moment (μ in μB), recovery time (τ in s, μs , and ns), and gas sensitivity (ε in%) for toxic gas absorbed on LNBP at different converge. The recovery time is considered with IR radiations at room temperature.

Type	E_{ads}	μ	τ	ε
NO_2 -LNBP	-2.55	0	$1.24\text{E}+31$ s	84.79 %
2NO_2 -LNBP	-1.12	0.94	$7.84\text{E}+6$ s	87.76 %
3NO_2 -LNBP	-0.29	1.00	6.97 ns	92.63 %
SO_2 -LNBP	-1.37	0.64	$1.50\text{E}+11$ s	70.81 %
2SO_2 -LNBP	-0.73	0.92	1.80 s	58.07 %
3SO_2 -LNBP	-0.24	0.98	33.3 ns	
H_2S -LNBP	-0.68	0	0.31 s	25.16 %
$2\text{H}_2\text{S}$ -LNBP	-0.89	0	918 s	39.77 %
$3\text{H}_2\text{S}$ -LNBP	-0.35	0	9.66 μs	13.49 %

produces only 0.03 e of charge transfer from the Li atom. The charge aggregation region is relatively sparse and there is hardly any electron supply between NBP and Li atom.

The TDOS and PDOS of the NO_2 -LNBP system adsorbed under a single gas molecule are given in Fig. 5(a, d). The TDOS obviously shifts toward higher energy levels, and the electron density changes in the conduction and valence bands compared to LNBP. It is evident from PDOS that orbital hybridization between O-2p and N-2p orbitals with P-3p orbitals after activation creates new peaks, which appear at approximately -1.4 eV and -6 eV in TDOS. Notably, a coupling peak is

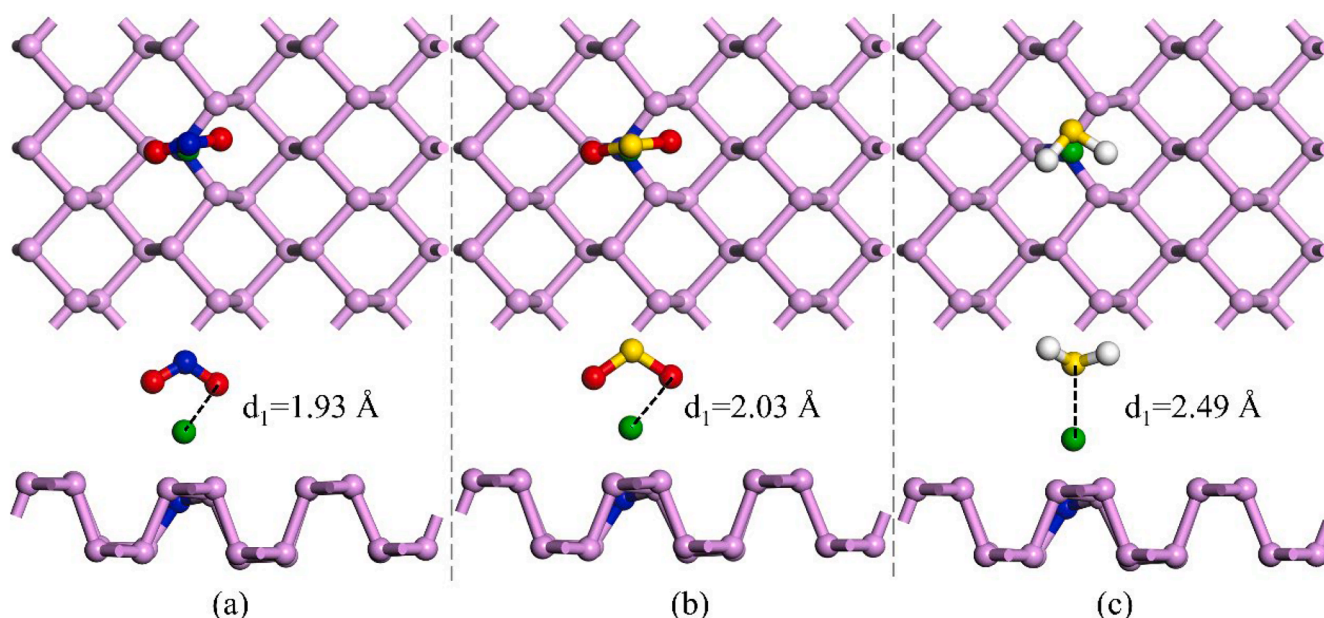


Fig. 3. The most stable adsorption structures in the NO_2 -LNBP, SO_2 -LNBP, and H_2S -LNBP systems. The d_1 represents the shortest distance from the gas to the Li atom.

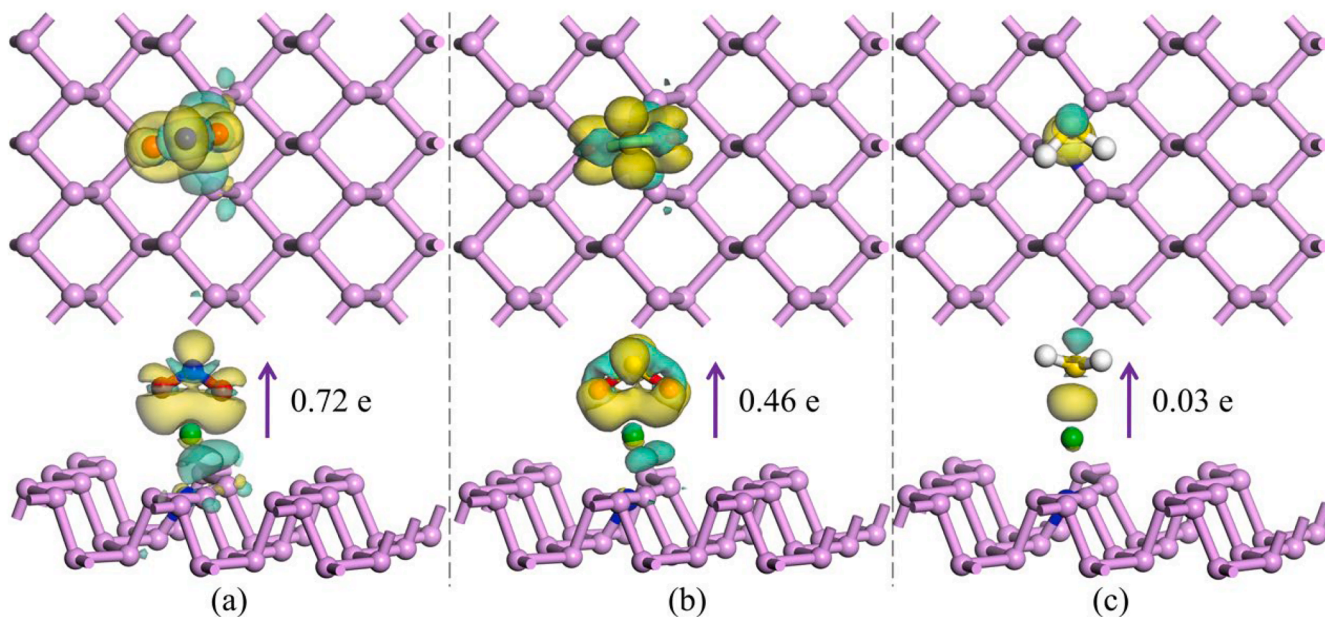


Fig. 4. Charge density difference in the NO₂-LNBP, SO₂-LNBP, and H₂S-LNBP systems, where the isosurfaces correspond to 0.015 e Å⁻³.

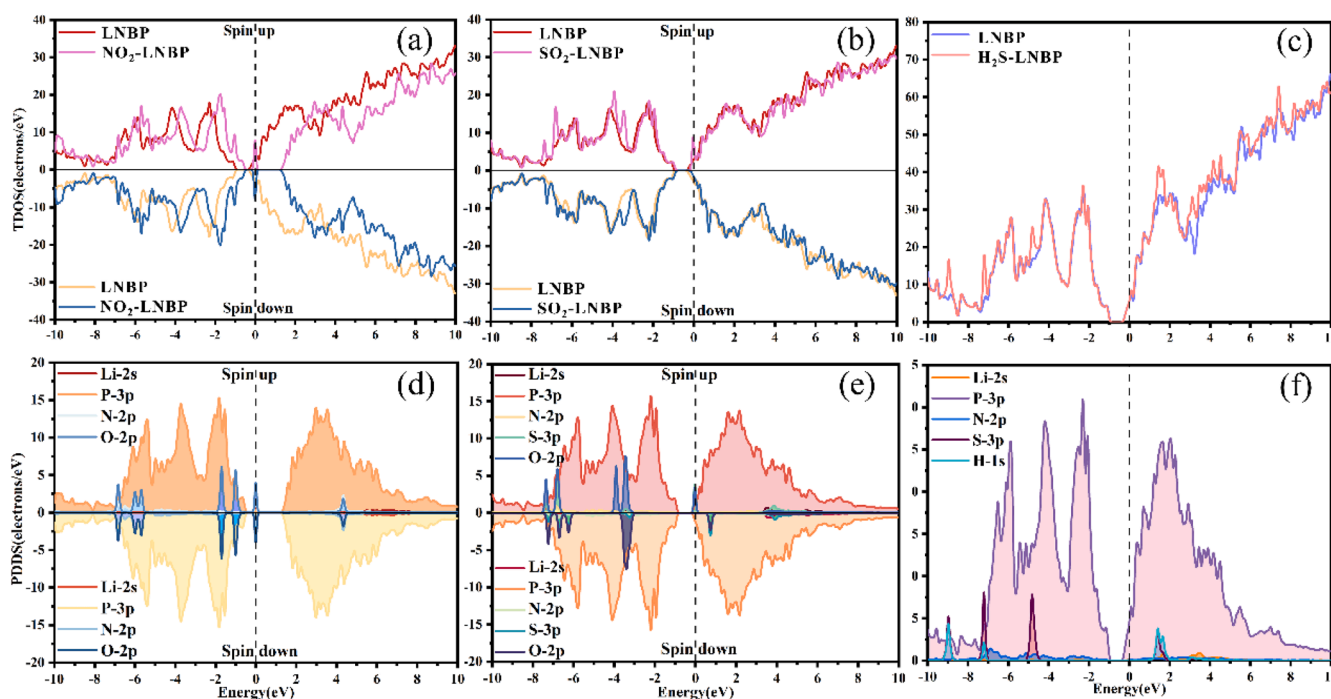


Fig. 5. TDOS and PDOS adsorbed by single gas molecule (a, d) NO₂, (b, e) SO₂, (c, f) H₂S on LNBP. The positive and negative DOS denote the spin-up and spin-down states, respectively. The vertical dashed line indicates the Fermi energy.

created close to the Fermi energy due to the hybridization of O-2p and P-3p orbitals and then induces an upward movement of the valence band, with an increase across the band gap by 1.43 eV. The Fermi energy of LNBP exhibits a downward shift and reverts to p-type conductivity. The N and O atoms introduce the symmetric states near VBM and CBM for the NO₂-LNBP adsorption system, respectively. After SO₂ is adsorbed on LNBP, there is a magnetic moment of 0.64 eV, indicating a spin-polarized system. The TDOS analysis from Fig. 5(b) reveals that the SO₂-LNBP system has a little alteration in the conduction band and changes below the Fermi energy similar to the NO₂-LNBP system, with peaks at approximately -3.7 eV, -4 eV, and -7 eV. Fig. 5(e) displays an

asymmetric distribution of peaks produced by the electrons in two spin-polarized directions and changes at these locations due to the hybridization of the SO₂ orbital with P-3p orbitals. The spin-up O-2p orbitals undergo coupling with P-3p orbitals, creating an impurity energy level and inducing a downshift of Fermi energy. Their hybridization is also created in the valence band within -3.7 eV and -7 eV, which relates to the large ionization energy of O. It is noteworthy that the Li-2s orbital hybridizes with S-3p orbitals around 3.6 eV. The TDOS and PDOS of H₂S adsorbed by LNBP are shown in Fig. 5(c, f). Several minimal changes are observed in TDOS after the adsorption. The total DOS is contributed by the P-3p orbitals, which have little interaction with other orbitals and

maintain semiconducting behavior. Unlike NO_2 and SO_2 , coupling peaks are created close to the Fermi energy, which directly indicates that the LNBP exhibits weak adsorption strength with H_2S , and this is consistent with previous analyses of LBP [38].

3.3. Effect of gas molecule coverage on the adsorption mechanism and sensing performance of toxic gases by LNBP

3.3.1. Effect of gas molecule coverage on the adsorption mechanism

To fit the actual situation in the chemical industry, the second and third gas molecules are put on the most stable adsorption structure to study the coverage effect on LNBP.

The optimized structures and calculated properties for the adsorption of NO_2 under the effect of coverage are shown in Fig. 6 and Fig. S8. In the adsorption systems in two-gas molecules and three-gas molecules, NO_2 molecules start being absorbed around the Li atom with E_{ads} of -1.12 eV and -0.29 eV, along with corresponding d_1 , d_2 , of 2.02 Å, 2.15 Å and d_1 , d_2 , d_3 of 2.46 Å, 2.57 Å, 2.81 Å. The adsorption distance of the Li atom increases significantly in the 3NO_2 -LNBP adsorption system, with a gradual decrease in adsorption energies, indicating a declining capacity to absorb NO_2 molecules effectively under the effect of coverage. In contrast, the adsorption process of NO_2 molecules produces higher charge transfer of 0.79 e and 0.95 e due to the increasing quantity of NO_2 molecules. As depicted in Fig. S8, each newly adsorbed NO_2 molecule creates a shared charge accumulation with the Li atom, showing an interaction between Li and gases. Both the adsorption positions of NO_2 and Li atoms vary with increasing coverage.

According to Lei et al. [37], the presence of nitrogen-based gases in the LBP system induces electronic spin-polarization. Thus, the consideration is introduced for NO_2 adsorption. In the NO_2 -LNBP system, the magnetic moment is zero, proving no spin polarization phenomena are generated. Whereas in the 2NO_2 -LNBP and 3NO_2 -LNBP systems, a magnetic moment of 0.94 μB and 1 μB , respectively, is produced in LNBP, which suggests spin polarization occurs as the coverage of NO_2 molecules increases. The spin-polarized TDOS and PDOS are shown in Fig. 6(b-f). There are some peaks created in the gap along a spin-down direction, inducing an increase of CBM in the 2NO_2 -LNBP and

3NO_2 -LNBP systems. The PDOS exhibits asymmetric distribution differences between spin-up and spin-down states. These distinctions are created due to the orbital coupling of N-2p and O-2p. The band gap of the NO_2 -LNBP system increases to 1.46 eV and 1.53 eV, as a spin-up peak near the Fermi energy caused by the hybridization of O-2p and P-3p orbitals and a spin-down peak at nearly 1.6 eV induced by the NO_2 orbital. Furthermore, the increasing coverage results in a more pronounced hybridization of NO_2 orbital and P-3p orbitals in the valence band both in spin-up and spin-down directions. In contrast, more electron density peaks are observed in the conduction band region for the spin-down state, indicating the stability of the adsorption decreases.

The adsorption structures and adsorption properties of the 2SO_2 -LNBP and 3SO_2 -LNBP systems are depicted in Fig. 7(a, d) and Table 3. and the E_{ads} are -0.74 eV and -0.24 eV. The corresponding d_1 , d_2 are 2.64 Å, 2.69 Å in the 2SO_2 -LNBP system, and d_1 , d_2 , d_3 are 2.65 Å, 2.67 Å, 3.97 Å in the 3SO_2 -LNBP system, respectively. The SO_2 molecules are effectively adsorbed around the Li atom in the 2SO_2 -LNBP system, with increasing adsorption distance, and render less capability for gas adsorption. While the third SO_2 molecule is 3.97 Å away from the Li atom in the 3SO_2 -LNBP system, indicating an unstable adsorption. As shown in Fig. S9, the charge transfer of SO_2 increases to 0.62 e in the 2SO_2 -LNBP system, but the phenomenon of being an acceptor is not clear on the SO_2 molecule further away from the Li atom, which creates a charge transfer of 0.65 e in the 3SO_2 -LNBP system. The difference (0.03 e) in charge transfer suggests the limitation of SO_2 adsorption capability on the LNBP in the 2SO_2 -LNBP system. From Table 3, the adsorption of SO_2 at different coverages produces a magnetic moment of 0.92 μB and 0.98 μB , respectively, indicating that the SO_2 -LNBP is always a magnetic system.

The Fig. 7(b-f) displays the TDOS and PDOS for the SO_2 -LNBP system. Noticeable differences are observed around 0 eV, -3.5 eV, -7 eV, and 0.5 eV from TDOS. The increase in the coverage of the SO_2 molecules makes the orbit hybridization of SO_2 and P-3p orbitals more pronounced to produce asymmetric electronic peaks in the presence of different spins close to the Fermi energy. Unlike the case in the single gas molecule adsorption system, the SO_2 orbital also shows a decrease in spin hybridization with Li-2 s orbital in the conduction band (4 – 5 eV).

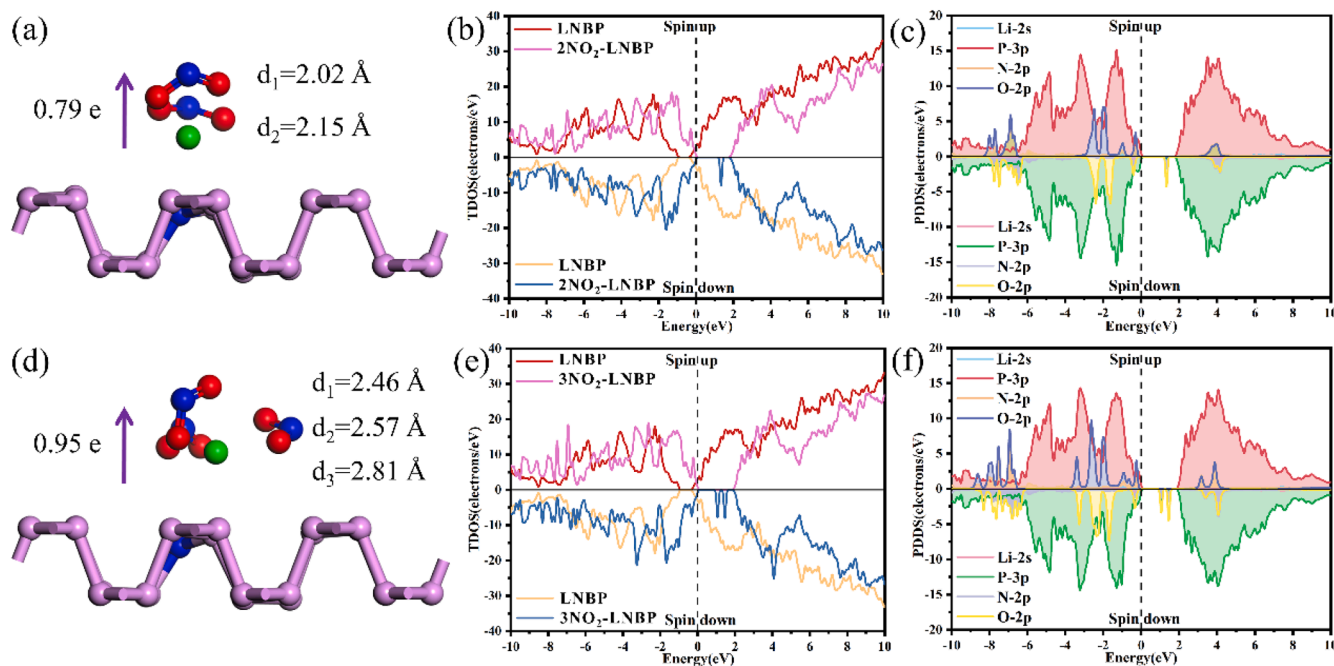


Fig. 6. Side views of the most stable adsorption for the 2NO_2 -LNBP and 3NO_2 -LNBP systems. The d_1 , d_2 and d_3 represent the shortest adsorption distance from every gas molecule to the Li atom. The spin-polarized TDOS of the 2NO_2 -LNBP system at (b) and the 3NO_2 -LNBP system at (e), where the corresponding PDOS (c, f) are also displayed.

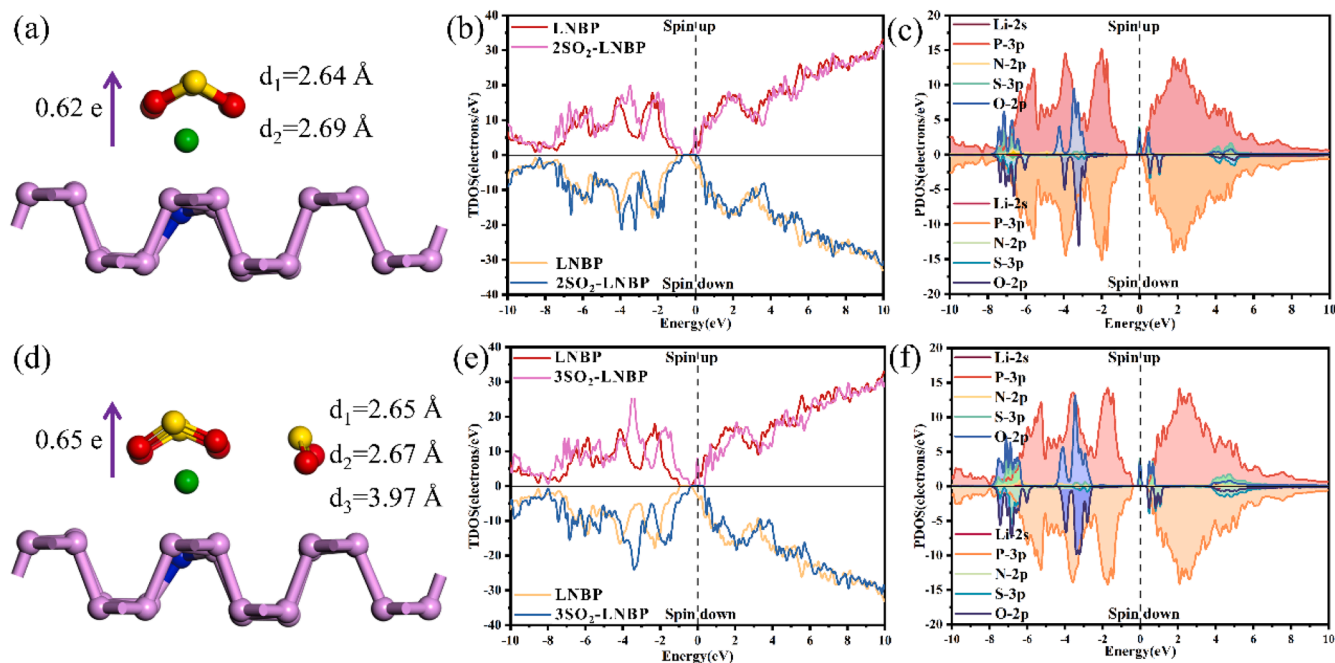


Fig. 7. (a, d) The most stable adsorption structures, (b, e) the TDOS and (c, f) the corresponding PDOS in the 2SO₂-LNBP and the 3SO₂-LNBP systems, respectively.

Due to the extra SO₂ adsorption, upward movements of the VBM and the spin-up peak close to the Fermi energy cause the disappearance of the band gap, indicating a transition from semiconductor to metal in the 3SO₂-LNBP system. The discretization of the DOS increases in the region of -6 eV to -8 eV, confirming that the system tends to be unstable in case of increased coverage.

As shown in Fig. 8(a, d) and Table 3, the E_{ads} of LNBP in the 2H₂S-LNBP and the 3H₂S-LNBP adsorption systems are -0.89 eV and -0.35 eV, accompanied by d₁, d₂ of 2.49 Å, 2.56 Å, and d₁, d₂, d₃ of 2.52 Å, 2.61 Å, 2.69 Å. There is an increase in adsorption energy of about 0.25 eV in the 2SO₂-LNBP system. The distances from H₂S to Li atom change, varying from -0.02 Å to 0.15 Å, and produce a total charge transfer of

0.17 e and 0.20 e in the 2H₂S-LNBP and 3H₂S-LNBP systems. From Fig. S10, more electrons form an aggregation in the region between the Li atom and H₂S molecules as the coverage increases, demonstrating that LNBP shows better adsorption for H₂S in the 2H₂S-LNBP system. However, the TDOS of the H₂S-LNBP system in both the CBM and VBM has only slightly changed compared to LNBP. PDOS analysis reveals a hybridization of the S-3p orbitals and H-1 s orbital at -8 eV with P-3p orbitals, while in the 3H₂S-LNBP system, there are fewer orbital hybridizations between H-1 s, S-3p, and P-3p, which weakens the interaction with the LNBP, offering an explanation for lower adsorption energy.

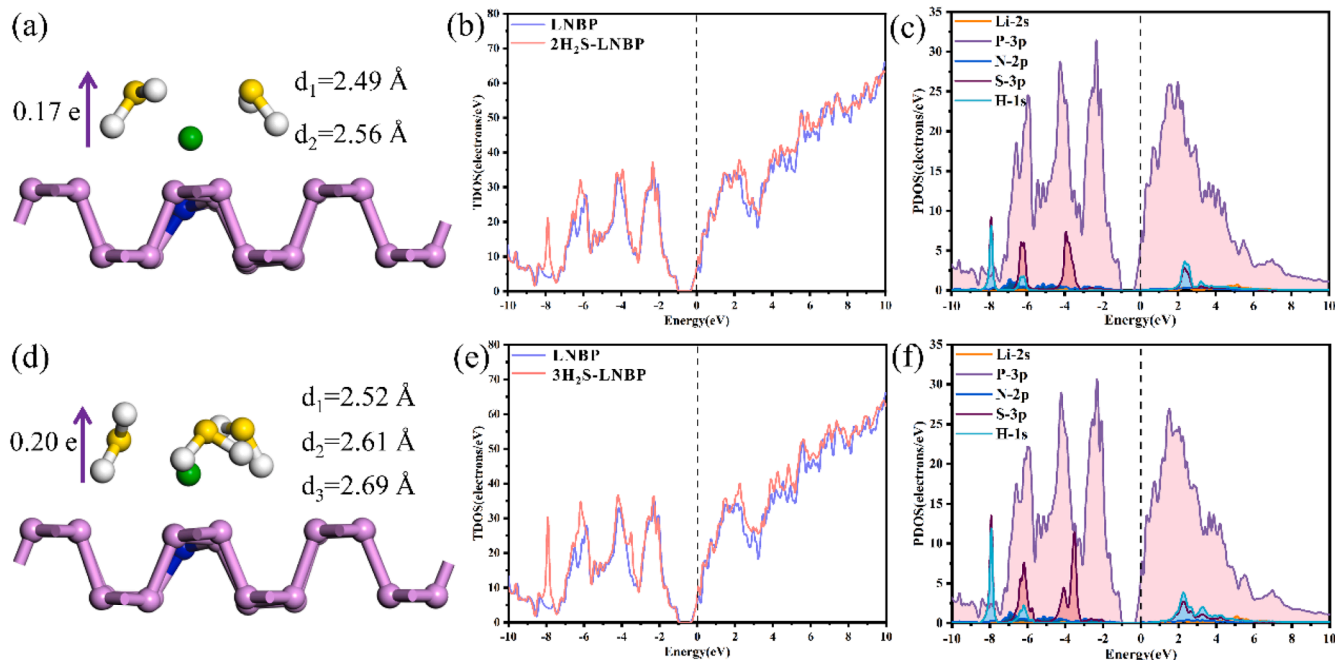


Fig. 8. (a, d) The most stable adsorption structures, (b, e) the TDOS and (c, f) the corresponding PDOS in the 2H₂S-LNBP and 3H₂S-LNBP systems, respectively.

3.3.2. Effect of gas molecule coverage on the sensing performance

It was known that the main parameters of a gas sensor are sensitivity, selectivity, and recovery time. The computational equations were given below [61,62]:

$$\varepsilon = \frac{|\sigma_g - \sigma_0|}{\sigma_0} \times 100\% \quad (4)$$

$$\tau = \nu_0^{-1} \exp\left(\frac{-\Delta E_{\text{ads}}}{k_B T}\right) \quad (5)$$

In which σ_g showed the electrical conductance of the system, and σ_0 represented the zero-bias conductivity of the LNBP before the adsorption; ν_0 denoted the attempt frequency (10^{12} s^{-1}), and ΔE_{ads} was the adsorption energy (eV) according to E_q . (3). The formula for the conductivity was calculated from the following equation [63]:

$$\sigma = CT^2 \exp\left(\frac{-E_{\text{gap}}}{2k_B T}\right) \quad (6)$$

Where T showed the operating temperature (298 K), k_B represented the Boltzmann constant ($8.62 \times 10^{-5} \text{ eV K}^{-1}$), and C signified the system-dependent constant of proportionality (electrons $\text{m}^{-3} \text{ K}^{-3/2}$).

To further understand the adsorption mechanism, the work function (Φ) was calculated. The Φ described the minimum energy required to displace an electron from the surface of the base material which was approximated as

$$\Phi = V(\Phi) - E_F \quad (7)$$

Where Φ , $V(\Phi)$, and E_F represented the work function, electrostatic potential at the vacuum level, and Fermi level, respectively.

As Table 3 demonstrates, the gas sensitivity and recovery time of three gases on LNBP are different. An effective gas monitor is built on the principle that the shorter the recovery time, the more proficient the sensor is at identifying and disengaging the gas. Remarkably, LNBP manifests a high sensitivity of 84.79 % and a recovery time of $1.24\text{E}+31$ s in the NO_2 -LNBP system, revealing its effectiveness in capturing NO_2 as an ideal adsorbent. The gas sensitivity is 87.76 % and 92.63 % in the 2NO_2 -LNBP and the 3NO_2 -LNBP systems, indicating enhanced sensing

properties due to the higher charge transfer and variable band gap value. It shows a recovery time of $7.84\text{E}+6$ s in the 2NO_2 -LNBP system, making it suitable for industrial use as a toxic gas detector. The recovery time is scaled down to 6.97 ns due to the decrease in adsorption energy in the 3NO_2 -LNBP system, resulting in the loss of performance of LNBP at high coverage. Besides, we analyze the work function (Φ) of LNBP in the pristine form as well as with the presence of adsorbed three molecules, as shown in Fig. 9. Tuning and optimization of Φ is important for making effective sensors and devices without ruining the electronic properties. After the Li decoration, the Φ decreases from 5.01 eV to 3.38 eV. This indicates that the decoration of the Li atom increases the carrier concentration and makes it easier to remove the electrons from the LNBP surface. Upon NO_2 adsorption, electron withdrawal from the LNBP lowers the Fermi level and increases the work function of the system. Accordingly, the adsorbed NO_2 induces large charge transfer from the LNBP and thus greatly enhances the work function to 5.46 eV, 5.06 eV, and 4.97 eV, respectively. The significant figure of merit indicates that LNBP is very sensitive to NO_2 and decreases with increasing coverage, consistent with the change in sensitivity.

For the adsorption of SO_2 , the sensing performance reveals that the sensitivity of SO_2 adsorbed on LNBP is 70.81 % with a recovery time of $1.5\text{E}+11$ s, which shows the ability to absorb SO_2 on its surface in the SO_2 -LNBP system stably. Furthermore, it has a sensitivity of 58.07 % in the 2SO_2 -LNBP system, with a recovery time of 1.8 s. Although the sensitivity decreases, it is suitable for the 2SO_2 -LNBP system as a gas sensor with a short recovery time, while in the 3SO_2 -LNBP system, the gas sensitivity losses and the recovery time is considerably reduced at 33.3 ns due to the disappearance of the band gap, which also reflects the unstable adsorption of 3SO_2 upon LNBP. With the change in coverage, the calculated Φ are 3.87 eV, 3.94 eV, and 4.10 eV. Relative to LNBP, there is still a significant increase of more than 0.5 eV after the adsorption of SO_2 . This variation suggests that the ability of electrons to move across the surface of the LNBP diminishes, which is consistent with the decrease in gas sensing with changes in coverage.

However, its performance for the H_2S -LNBP system where the recovery time is 0.31 s and has a sensitivity of 25.16 %. Furthermore, in the $2\text{H}_2\text{S}$ -LNBP system, the recovery time increases to 918 s, and the

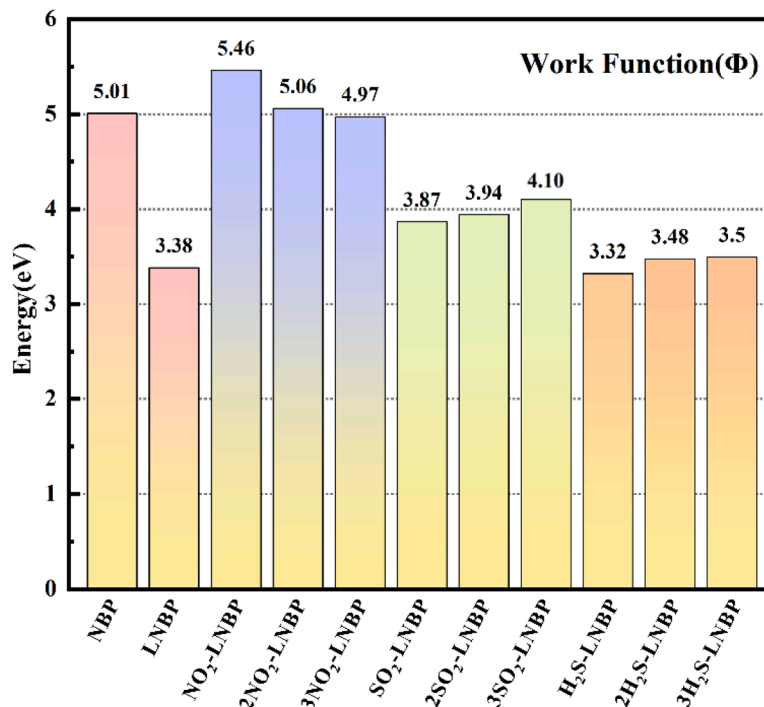


Fig. 9. The calculated work function (Φ) of NO_2 , SO_2 , and H_2S gas molecules adsorbed on LNBP and LNBP.

sensing rate of 39.77 % is achieved, indicating that LNBP will perform better sensing applications at appropriately high concentrations, while it hardly displays the gas adsorption and sensing performance of LNBP with a sensitivity and recovery time of 13.49 % and 9.66 μ s in the 3H₂S-LNBP system. Finally, due to the weak interaction, the H₂S molecules are associated with less charge transfer and only increase the Φ in 3.32 eV to 3.49 eV slightly, confirming the low sensitivity with subtle amplitude changes in the power function.

3.4. Toxic gas (NO₂, SO₂, H₂S) dissociation performances of LNBP

Based on the most stable structures in the case of adsorbing one gas molecule, the dissociation process of NO₂, SO₂, and H₂S on LNBP is studied. A transition state through a minimum energy path is examined based on the complete linear synchronous transit (LST), and the quadratic synchronous transit (QST) method is used to follow the reaction path of the reaction between LNBP and the gas molecules [64].

The geometries of the initial state (IS), transition state (TS), and final state (FS) of toxic gases are depicted in Fig. 10. The dissociation process of NO₂ displayed in Fig. 11(a) is analyzed as NO₂ → NO + O. Following the breaking of the bond with the N atom in TS, the dissociated O atom creates a covalent interaction with the P atom. Due to its high electronegativity, the N atom attracts higher than O and approaches the Li atom at an adsorption distance of 2.06 Å, causing the NO fragment to rotate at a small angle. A bonding connection is formed at a distance of 1.54 Å when a P atom near the doped N atom desorbs the O atom in the FS. The NO fragment is adsorbed by Li atom at a distance of 1.91 Å, transiting from near-parallel to perpendicular alignment. The energy barrier to be overcome in this reaction process is 1.24 eV and the energy of the final state is −0.47 eV, suggesting that the exothermic process should ideally be conducted under higher temperature. The low activation energy exhibited by the reactants demonstrates their potential competitiveness in addressing exhaust gas challenges [39].

The dissociation process of SO₂ on LNBP is depicted in Fig. 11(b). It can be found that SO₂ undergoes a reaction: SO₂ → SO + O. Upon rotation of the entire molecule at a certain angle, one of the O atoms breaks its bond with the S atom and is adsorbed by a neighboring P atom adjacent to the N atom, forming a bonding distance of 2.14 Å. However, the SO fragment remains at an adsorption distance of 2.15 Å with the Li atom. With regard to FS, both SO fragment and O atom exhibit strong interaction with P atoms with a bond length of 2.15 Å and 1.54 Å. The longer adsorption distance (3.20 Å) between the Li atom and NBP demonstrates that this dissociation process potentially damages the sensor LNBP, which is more likely to dissociate SO₂ on the BP surface. As

the reaction proceeds, it becomes evident that the dissociation energy of SO₂ closely resembles that of NO₂, with a higher energy barrier of 1.36 eV to be overcome during this reaction. The energy of the FS is −0.42 eV, indicating the possibility that the reaction would happen at a higher temperature.

For H₂S (Fig. 11(c)), the TS is followed by the reaction: H₂S → HS + H. An H atom in H₂S breaks its bond with S and gradually moves closely to LNBP. The residual HS fragment remains significantly distant from the Li atom during adsorption. In the FS, the dissociated H atom is adsorbed by adjacent P atoms near the N atom, with a bond length of 1.62 Å. In addition, a higher energy barrier of 2.71 eV makes the reaction impossible. The energy of the FS is 1.10 eV, thereby indicating a heat-absorbing reaction. The combination of the above also suggests that it is almost impossible for H₂S to dissociate at LNBP in an industrial setting. An alternative explanation is attributed to the weak interaction between H₂S and LNBP, hindering further dissociation. According to the previous chemical reaction on 2D materials [65], it is found that LNBP plays an excellent sensing performance for toxic gases at a certain coverage, as well as its effectiveness in removing NO₂ and SO₂ from exhaust gas treatment processes conducted at elevated temperatures within certain industrial environments.

4. Conclusion

In summary, the DFT calculations are adapted to provide significant insights into the adsorption in one-, two-, and three-gas molecule systems behavior of toxic gases on LNBP. The results reveal that the interaction between the Li atom and NBP is due to the hybridization of N-2p with Li-2 s orbital and a narrower range of charge transfer, introducing larger adsorption energy of −3.56 eV compared to LBP. The intense orbital hybridization between O-2p and P-3p near the Fermi energy and large charge accumulation are created when capturing NO₂ and SO₂ molecules, demonstrating significant interaction and selectivity compared to that of H₂S. With regard to NO₂ on LNBP, a high sensitivity of up to 92.6 % and recovery time from 1.24E+31 s to 6.97 ps are changed as the coverage increases. Whereas a magnetic moment of 0.94 μ B and 1 μ B are produced in the 2NO₂-LNBP and 3NO₂-LNBP systems, respectively, which suggests that the spin polarization occurs as the coverage of NO₂ molecules increases. Besides, it achieves an effective detection for SO₂ adsorbed on LNBP due to the suitable recovery time at certain coverage. Meanwhile, the 3SO₂-LNBP system has transitioned from semiconductor to metal in terms of electronic properties due to the extra molecule adsorption. The observed significant changes in the work function after the adsorption have a signature of high sensitivity and

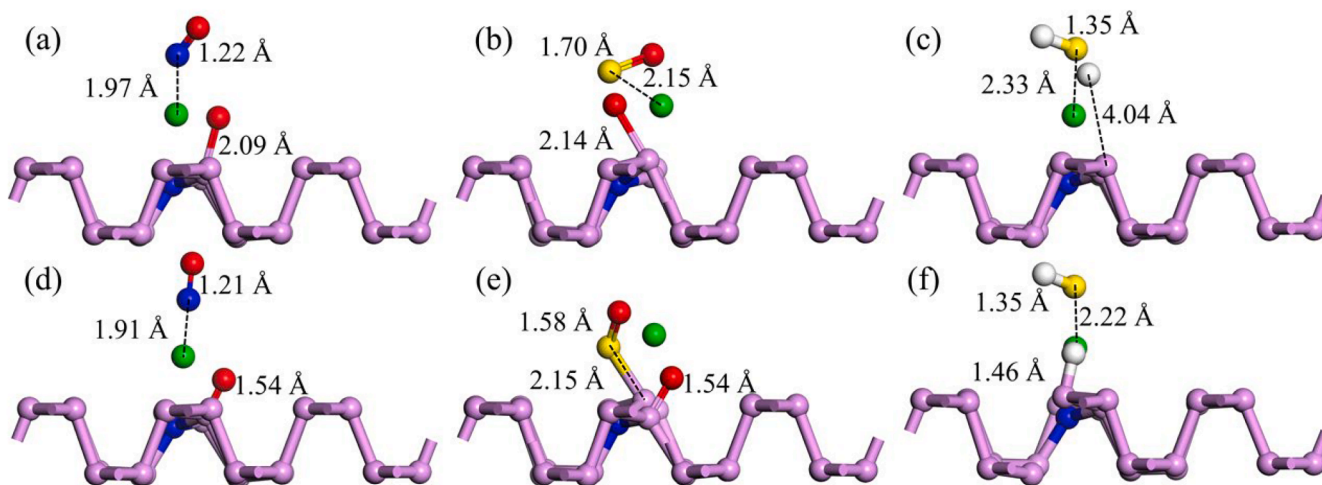


Fig. 10. The transition states of (a) NO₂, (b) SO₂, and (c) H₂S fragments on LNBP, and their final states are represented in (d-f). The values are the bond lengths and the distance between two atoms.

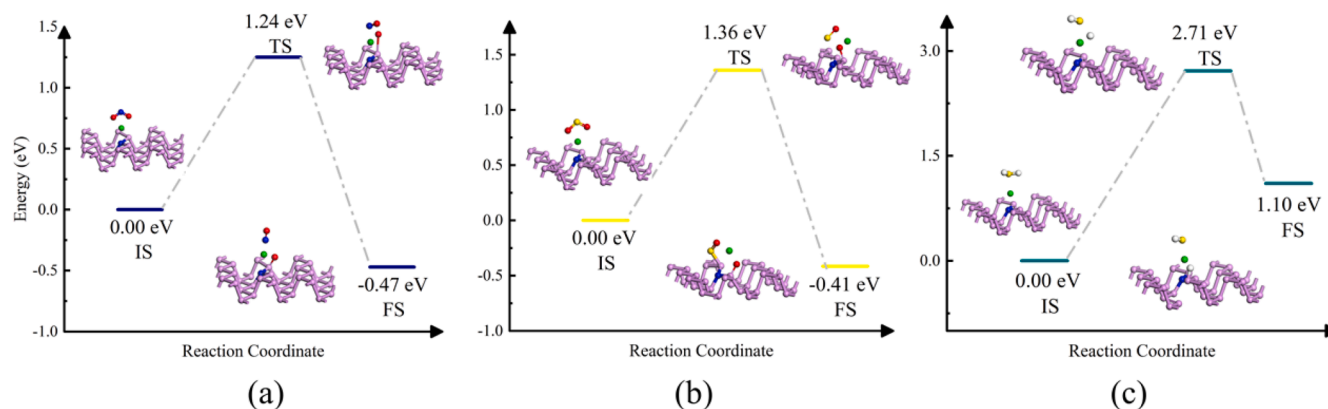


Fig. 11. Energy profiles for toxic gases ((a) NO₂, (b) SO₂, (c) H₂S) dissociation on LNBP. The energy value over TS is the reaction energy barrier.

selectivity of the LNBP, following the other: NO₂>SO₂>H₂S. Moreover, a good catalytic ability for the NO₂-LNBP system is discussed, which has the potential to achieve spontaneous dissociation of NO₂ at high temperatures. Our theoretical studies suggest that it is essential to explore the optimal environment for detecting and adsorbing toxic gas by LNBP in practical application.

CRediT authorship contribution statement

Zilin Chen: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Ruolan Li:** Writing – original draft, Visualization, Investigation, Formal analysis. **Xin Peng:** Writing – original draft, Investigation, Data curation. **Hao Jiang:** Validation, Supervision. **Huadong Zeng:** Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests and personal relationships which may be considered as potential competing interests: Huadong Zeng reports financial support was provided by South China Normal University.

Data availability

Data will be made available on request.

Acknowledgments

We thank the South China Normal University, for its assistance with the Material Studio simulations. This work was supported by South China Normal University Innovation Training Program for University Students (No. 202280001) and South China Normal University Young Teachers Research and Cultivation Fund Project (21KJ10).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.surfin.2024.104872](https://doi.org/10.1016/j.surfin.2024.104872).

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